

CSC1018 - Logic

Propositional Logic

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Propositions

Propositional Logic is a logical system upon which other logical systems are constructed.

A *proposition* is a atomic statement that either has the truth value *true* or *false*.

Some example propositions:

- $1 + 1 = 2$
- $1 + 2 = 4$
- The Earth is closer to the Sun than Mars.
- My eyes are green.

Boolean Operators

Negation

Negation is a unary boolean operator. *Unary* means it takes one argument. *Boolean* means it evaluates to either *true* or *false*.

The truth table for *negation* is:

x	$\neg x$
<i>false</i>	<i>true</i>
<i>true</i>	<i>false</i>

$\neg x$ can be read as “not x ”

Boolean Operators (2)

Conjunction

Conjunction is a binary boolean operator. *Binary* means it takes two argument.

- Binary operators are used to combine two propositions into a propositional formula.
- A proposition by itself is also a propositional formula.

The truth table for *conjunction* is:

x	y	$x \wedge y$
<i>false</i>	<i>false</i>	<i>false</i>
<i>true</i>	<i>false</i>	<i>false</i>
<i>false</i>	<i>true</i>	<i>false</i>
<i>true</i>	<i>true</i>	<i>true</i>

$x \wedge y$ can be read as “ x and y ”

Boolean Operators (3)

Disjunction

Disjunction is a binary boolean operator.

The truth table for *disjunction* is:

x	y	$x \vee y$
<i>false</i>	<i>false</i>	<i>false</i>
<i>true</i>	<i>false</i>	<i>true</i>
<i>false</i>	<i>true</i>	<i>true</i>
<i>true</i>	<i>true</i>	<i>true</i>

$x \vee y$ can be read as “x or y”

Boolean Operators (4)

Exclusive-or

Exclusive-or is a binary boolean operator. It is related to the *disjunction* operator but more closely matches the “or” in natural language, e.g. “I can go home by bus or by car” but not both at the **same** time.

The truth table for *exclusive-or* is:

x	y	$x \oplus y$
<i>false</i>	<i>false</i>	<i>false</i>
<i>true</i>	<i>false</i>	<i>true</i>
<i>false</i>	<i>true</i>	<i>true</i>
<i>true</i>	<i>true</i>	<i>false</i>

Exclusive-or is “reversible”, i.e. $(x \oplus y) \oplus y = x$

Boolean Operators (5)

Implication

Implication is a binary boolean operator.

The truth table for *implication* is:

x	y	$x \rightarrow y$
<i>false</i>	<i>false</i>	<i>true</i>
<i>true</i>	<i>false</i>	<i>false</i>
<i>false</i>	<i>true</i>	<i>true</i>
<i>true</i>	<i>true</i>	<i>true</i>

$x \rightarrow y$ can be read as “ x materially implies y ” (or “ x implies y ”)

In $x \rightarrow y$, x is the *antecedent* and y is the *consequent*. Note that *material implication* does not mean that x causes y . Only that $x \rightarrow y$ is *false* if x is *true* and y is *false*.

Boolean Operators (6)

Equivalence

Equivalence is a binary boolean operator.

The truth table for *equivalence* is:

x	y	$x \leftrightarrow y$
<i>false</i>	<i>false</i>	<i>true</i>
<i>true</i>	<i>false</i>	<i>false</i>
<i>false</i>	<i>true</i>	<i>false</i>
<i>true</i>	<i>true</i>	<i>true</i>

Boolean Operators (7)

Nand and Nor

Both *Nand* and *Nor* are binary boolean operators.

The truth table for *nand* is: The truth table for *nor* is:

x	y	$x \uparrow y$
<i>false</i>	<i>false</i>	<i>true</i>
<i>true</i>	<i>false</i>	<i>true</i>
<i>false</i>	<i>true</i>	<i>true</i>
<i>true</i>	<i>true</i>	<i>false</i>

x	y	$x \downarrow y$
<i>false</i>	<i>false</i>	<i>true</i>
<i>true</i>	<i>false</i>	<i>false</i>
<i>false</i>	<i>true</i>	<i>false</i>
<i>true</i>	<i>true</i>	<i>false</i>

Nand and *nor* are the negation of *conjunction* and *disjunction*. They are important because a combination of either of them individually can implement all other boolean operators.

Boolean Formulae

If $\mathcal{P}$ is the set of atomic propositions, a Boolean formula bf can be derived from the following rules:

$$bf ::= p \text{ where } p \in \mathcal{P}$$

$$bf ::= \neg bf$$

$$bf ::= bf \text{ op } bf$$

$$op ::= \wedge \mid \vee \mid \oplus \mid \rightarrow \mid \leftrightarrow \mid \uparrow \mid \downarrow$$

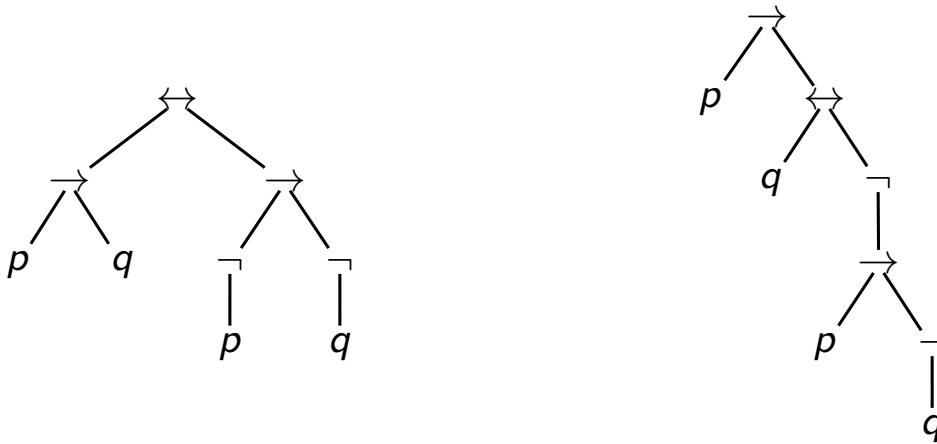
We will denote the set of boolean formulae derived by the above rule as $\mathcal{F}$.

Ambiguity, Precedence and Associativity

The previous rules by themselves are not sufficient to assign a single meaning to the following formula.

$$p \rightarrow q \leftrightarrow \neg p \rightarrow \neg q$$

We can represent this formula by at least two different trees.



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Ambiguity, Precedence and Associativity (2)

When the same formula can have more than one meaning, the formula is *ambiguous*.

This problem can be addressed by:

- using parentheses,
- assigning a *precedence* to the operators, and
- assuming that operators *associate* to the right.

The order of *precedence* from highest to lowest is:

$$\neg$$

$$\wedge, \uparrow$$

$$\vee, \downarrow$$

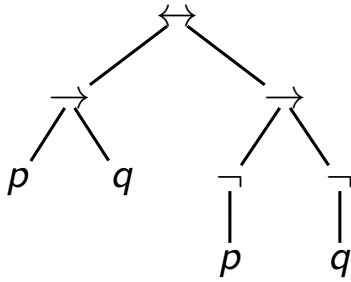
$$\rightarrow$$

$$\leftrightarrow, \oplus$$

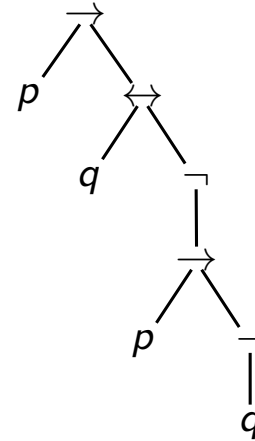
Ambiguity, Precedence and Associativity (3)

So, now, we have an only one meaning for each of the following formulae.

$$p \rightarrow q \leftrightarrow \neg p \rightarrow \neg q$$



$$p \rightarrow (q \leftrightarrow \neg(p \rightarrow \neg q))$$



Interpretations

An *interpretation* brings meaning (*semantics*) to a boolean formula. It allows us to evaluate the truth value of a formula.

Let $A \in \mathcal{F}$ be a formula and $\mathcal{P}_A$ be the set of atomic propositions appearing in A . An *interpretation* for A is a total function $\mathcal{I}_A : \mathcal{P}_A \mapsto \{\text{true}, \text{false}\}$ that assigns one of the truth value, *true* or *false*, to each atomic proposition in $\mathcal{P}_A$.

Let $\mathcal{I}_A$ be an interpretation of $A \in \mathcal{F}$, $v_{\mathcal{I}_A}(A)$, the *truth value of A under $\mathcal{I}_A$* is defined inductively on the structure of A .

$v_{\mathcal{I}_A}(A) = \mathcal{I}_A(A)$	if A is an atom	
$v_{\mathcal{I}_A}(\neg A) = \text{true}$	if $v_{\mathcal{I}_A}(A) = \text{false}$	$v_{\mathcal{I}_A}(\neg A) = \text{false}$ if $v_{\mathcal{I}_A}(A) = \text{true}$
$v_{\mathcal{I}_A}(A_1 \wedge A_2) = \text{true}$	if $v_{\mathcal{I}_A}(A_1) = \text{true}$ and $v_{\mathcal{I}_A}(A_2) = \text{true}$	$v_{\mathcal{I}_A}(A_1 \wedge A_2) = \text{false}$ otherwise
$v_{\mathcal{I}_A}(A_1 \vee A_2) = \text{true}$	if $v_{\mathcal{I}_A}(A_1) = \text{true}$ or $v_{\mathcal{I}_A}(A_2) = \text{true}$	$v_{\mathcal{I}_A}(A_1 \vee A_2) = \text{false}$ otherwise
$v_{\mathcal{I}_A}(A_1 \oplus A_2) = \text{false}$	if $v_{\mathcal{I}_A}(A_1) = v_{\mathcal{I}_A}(A_2)$	$v_{\mathcal{I}_A}(A_1 \oplus A_2) = \text{true}$ if $v_{\mathcal{I}_A}(A_1) \neq v_{\mathcal{I}_A}(A_2)$
$v_{\mathcal{I}_A}(A_1 \rightarrow A_2) = \text{false}$	if $v_{\mathcal{I}_A}(A_1) = \text{true}$ and $v_{\mathcal{I}_A}(A_2) = \text{false}$	$v_{\mathcal{I}_A}(A_1 \rightarrow A_2) = \text{true}$ otherwise
$v_{\mathcal{I}_A}(A_1 \leftrightarrow A_2) = \text{true}$	if $v_{\mathcal{I}_A}(A_1) = v_{\mathcal{I}_A}(A_2)$	$v_{\mathcal{I}_A}(A_1 \leftrightarrow A_2) = \text{false}$ if $v_{\mathcal{I}_A}(A_1) \neq v_{\mathcal{I}_A}(A_2)$
$v_{\mathcal{I}_A}(A_1 \uparrow A_2) = \text{false}$	if $v_{\mathcal{I}_A}(A_1) = \text{true}$ and $v_{\mathcal{I}_A}(A_2) = \text{true}$	$v_{\mathcal{I}_A}(A_1 \uparrow A_2) = \text{true}$ otherwise
$v_{\mathcal{I}_A}(A_1 \downarrow A_2) = \text{true}$	if $v_{\mathcal{I}_A}(A_1) = \text{false}$ and $v_{\mathcal{I}_A}(A_2) = \text{false}$	$v_{\mathcal{I}_A}(A_1 \downarrow A_2) = \text{false}$ otherwise

Interpretations (2)

Example:

$A = p \rightarrow q \leftrightarrow \neg q \rightarrow \neg p$ and $\mathcal{I}_A(p) = \text{false}$, $\mathcal{I}_A(q) = \text{true}$

$$v_{\mathcal{I}}(p) = \mathcal{I}_A(p) = \text{false}$$

$$v_{\mathcal{I}}(q) = \mathcal{I}_A(q) = \text{true}$$

$$v_{\mathcal{I}}(p \rightarrow q) = \text{true}$$

$$v_{\mathcal{I}}(\neg p) = \text{true}$$

$$v_{\mathcal{I}}(\neg q) = \text{false}$$

$$v_{\mathcal{I}}(\neg q \rightarrow \neg p) = \text{true}$$

$$v_{\mathcal{I}}(p \rightarrow q \leftrightarrow \neg p \rightarrow \neg q) = \text{true}$$

Interpretations (3)

Or we can use a *truth table*

p	q	$p \rightarrow q$	$\neg p$	$\neg q$	$\neg q \rightarrow \neg p$	$p \rightarrow q \leftrightarrow \neg q \rightarrow \neg p$
true	true	true	false	false	true	true
true	false	false	false	true	false	true
false	true	true	true	false	true	true
false	false	true	true	true	true	true

The interpretation we are looking for is

$\mathcal{I}_A(p) = \text{false}$, $\mathcal{I}_A(q) = \text{true}$.

The *truth table* evaluate the formula for all interpretations.

Let $A_1, A_2 \in \mathcal{F}$. if $v_{\mathcal{I}}(A_1) = v_{\mathcal{I}}(A_2)$ for all interpretations then A_1 is *logically equivalent* to A_2 , $A_1 \equiv A_2$.

Implication, $\rightarrow$, is different from logical equivalence, $\equiv$. $\rightarrow$ is a boolean operator whereas $\equiv$ describes a property of pairs of formulae.

Logically Equivalent Formulae

Absorption of Constants

$$A \vee \text{true} \equiv \text{true}$$

$$A \vee \text{false} \equiv A$$

$$A \rightarrow \text{true} \equiv \text{true}$$

$$A \rightarrow \text{false} \equiv \neg A$$

$$A \leftrightarrow \text{true} \equiv A$$

$$A \leftrightarrow \text{false} \equiv \neg A$$

$$A \wedge \text{true} \equiv A$$

$$A \wedge \text{false} \equiv \text{false}$$

$$\text{true} \rightarrow A \equiv A$$

$$\text{false} \rightarrow A \equiv \text{true}$$

$$A \oplus \text{true} \equiv \neg A$$

$$A \oplus \text{false} \equiv A$$

Identical Operands

$$\neg\neg A \equiv A$$

$$A \wedge A \equiv A$$

$$A \vee \neg A \equiv \text{true}$$

$$A \rightarrow A \equiv \text{true}$$

$$A \leftrightarrow A \equiv \text{true}$$

$$A \uparrow A \equiv \neg A$$

$$A \vee A \equiv A$$

$$A \wedge \neg A \equiv \text{false}$$

$$A \oplus A \equiv \text{false}$$

$$A \downarrow A \equiv \neg A$$

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Logically Equivalent Formulae (2)

Commutativity

$$A \vee B \equiv B \vee A$$

$$A \leftrightarrow B \equiv B \leftrightarrow A$$

$$A \uparrow B \equiv B \uparrow A$$

$$A \wedge B \equiv B \wedge A$$

$$A \oplus B \equiv B \oplus A$$

$$A \downarrow B \equiv B \downarrow A$$

Implication is not commutative

$$A \rightarrow B \equiv \neg B \rightarrow \neg A$$

Associativity

$$A \vee (B \vee C) \equiv (A \vee B) \vee C$$

$$A \leftrightarrow (B \leftrightarrow C) \equiv (A \leftrightarrow B) \leftrightarrow C$$

$$A \wedge (B \wedge C) \equiv (A \wedge B) \wedge C$$

$$A \oplus (B \oplus C) \equiv (A \oplus B) \oplus C$$

Distributivity

$$A \vee (B \wedge C) \equiv (A \vee B) \wedge (A \vee C)$$

$$A \wedge (B \vee C) \equiv (A \wedge B) \vee (A \wedge C)$$

Logically Equivalent Formulae (3)

We can define boolean operators in terms of each other.

$$\begin{array}{ll}
 A \leftrightarrow B & \equiv (A \rightarrow B) \wedge (B \rightarrow A) & A \oplus B & \equiv \neg(A \rightarrow B) \vee \neg(B \rightarrow A) \\
 A \rightarrow B & \equiv \neg A \vee B & A \rightarrow B & \equiv \neg(A \wedge \neg B) \\
 A \vee B & \equiv \neg(\neg A \wedge \neg B) & A \wedge B & \equiv \neg(\neg A \vee \neg B) \\
 A \vee B & \equiv \neg A \rightarrow B & A \wedge B & \equiv \neg(A \rightarrow \neg B)
 \end{array}$$

What is the smallest set of boolean operators that can be used to define the remaining boolean operators?

The operators $\wedge, \vee, \oplus, \rightarrow, \leftrightarrow, \uparrow$ and $\downarrow$ can be defined in terms of $\neg$ and one of $\wedge, \vee$ or $\rightarrow$.

But since $A \uparrow A \equiv \neg A$ and $\neg(A \uparrow B) \equiv A \wedge B$, then $\uparrow$ is sufficient!

Similarly for $\downarrow$.

Substitution

A is a *subformula* of B if the tree that represents A is a subtree of the tree that represents B .

For example, $\neg p \rightarrow \neg q$ is a subformula of $p \rightarrow q \leftrightarrow \neg p \rightarrow \neg q$.

Let A be a subformula of B and A' be any formula. Then, $B\{A \leftarrow A'\}$, the *substitution* of A' for A in B , is the formula you get where every occurrence of A in B is replaced by A' .

If A is a subformula of B and $A \equiv A'$, then $B \equiv B\{A \leftarrow A'\}$.

Satisfiability and Validity

If $A \in \mathcal{F}$ then

- A is *satisfiable* iff $v_{\mathcal{I}}(A) = \text{true}$ for some interpretation $\mathcal{I}$.
 - A satisfying interpretation is a *model* of A .
- A is *valid*, denoted $\models A$, iff $v_{\mathcal{I}}(A) = \text{true}$ for all interpretations $\mathcal{I}$.
 - A valid formula is called a *tautology*.
- A is *unsatisfiable* iff it is not satisfiable.
- A is *falsifiable*, denoted $\not\models A$, iff it is not valid.

Let $A \in \mathcal{F}$. Then A is valid iff $\neg A$ is unsatisfiable, and A is satisfiable iff $\neg A$ is falsifiable.